

Experiment 7 : Sequence-to-Sequence Learning with Transformers

Objective :

The goal of this assignment is to implement a sequence-to-sequence (seq2seq) model for machine translation from English to Spanish.

Unlike traditional LSTM-based models, the Transformer architecture relies entirely on self-attention mechanisms and parallel computation. In this assignment, using PyTorch, you will:

- Implement the core components of the Transformer architecture (Encoder-Decoder structure).
- Understand and implement Self-Attention and Multi-Head Attention mechanisms.
- Implement Positional Encoding to incorporate sequence order information.
- Use learned sequence embeddings.
- Train and evaluate the model on an English-Spanish translation dataset using BLEU score.

Dataset :

You are provided with a text file containing English-Spanish sentence pairs, one pair per line,
separated by a tab:
Hello.

Hola.

How are you? ¿C'omo est'as?

I am fine. Estoy bien.

...

Note:

If the dataset is too large to train in reasonable time on your system, you may sample a subset

(e.g., 10,000 sentence pairs), and split it into:

- 80% training
- 10% validation
- 10% testing

Ensure all preprocessing (tokenization, lowercasing, padding, etc.) is properly handled.

Task :

Implement a complete Transformer-based Encoder-Decoder model for English-to-Spanish

translation. Your implementation must include:

1. Embedding Layer
 - Learn word embeddings for both source and target vocabularies.
2. Positional Encoding
 - Implement sinusoidal positional encoding and add it to input embeddings.
3. Multi-Head Self-Attention
 - Implement scaled dot-product attention.
 - Extend it to multi-head attention.
 - Include masking for padded tokens.
4. Transformer Encoder
 - Multi-head self-attention layer
 - Position-wise Feed Forward Network
 - Residual connections and Layer Normalization

- Stack multiple encoder layers

5. Transformer Decoder

- Masked multi-head self-attention
- Encoder-Decoder (cross) attention
- Feed Forward Network
- Residual connections and Layer Normalization
- Stack multiple decoder layers

6. Training

- Use teacher forcing during training.
- Use appropriate loss function (e.g., Cross-Entropy with padding mask).
- Train using an optimizer of your choice (e.g., Adam).

7. Evaluation

- Evaluate translation performance using BLEU score.
- Compare results with your previous LSTM-based models.
- Report training time and performance metrics.